LECTURE NOTES: SYNTHETIC CATEGORY THEORY

Taught in parallel at Universität Hamburg and Universität Regensburg

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Introduction

This course is an introduction to ∞-*category theory*. ∞-Categories are homotopical generalisations of ordinary categories. They provide a convenient language for organising mathematical objects and their relationships, and for formulating and proving theorems. Over the past few decades, they have become an important tool in areas of algebraic topology, algebraic geometry, number theory, homological algebra, and mathematical physics.

What is an ∞-category?

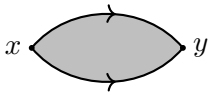
Informally, an ∞-category consists of

- objects $x, y, \ldots$;
- morphisms $f\colon x \to y$ between objects;
- 2-morphisms $x \overset{f}{\underset{g}{\Downarrow \alpha}} y$ between parallel morphisms
- 3-morphisms between parallel 2-morphisms;
- and so on;

such that all k-morphisms for $k \geq 2$ are invertible, in a suitably weak sense. Composition is associative and unital up to higher invertible morphisms, which themselves satisfy compatibility conditions in all higher dimensions.

A motivating example comes from topology. Given a topological space X, consider the ∞-category whose:

- objects are points of X;
- morphisms are paths between points in X; 

- 2-morphisms are homotopies between paths, fixing the endpoints; 

- 3-morphisms are homotopies between homotopies, fixing the boundaries;
- and so on.

Composition is given by concatenating paths and homotopies. Concatenation of paths is associative and unital up to homotopy: different ways of composing the same paths need not give exactly the same parametrised path, but they can be continuously deformed into one another. Moreover, reversing a path gives an inverse up to homotopy. Thus, in this example, even the morphisms are invertible in the appropriate sense. Such a structure is called an ∞-*groupoid*.

Axiomatic approach to ∞-categories

Traditionally, ∞-categories have been studied through concrete *models*, such as quasi-categories [Joy08, Lur09] and complete Segal spaces [Rez01, Ras23]. These models make the informal description above precise using constructions within set-theoretic foundations. Encoding ∞-categories in set-theoretic terms can make them difficult to work with: questions about intrinsic categorical properties may become hard combinatorial problems involving the chosen model.

In this course, we will take an axiomatic approach to ∞-category theory. Our approach differs from the model-based one described above in the following aspects: the foundations, including the structure identity principle, and the synthetic approach.

Homotopy type theory. The first aspect is a change of foundations. In ordinary set-based mathematics, given a set X and elements $x, y \in X$, the expression $x = y$ is a proposition: it is either true or false, and carries no further structure describing an identification between x and y.

In *homotopy type theory*, our basic objects are types. We write $x : X$ to say that x is a term, or element, of the type X. Given terms $x, y : X$, the expression

$$x = y$$

is itself a type, called an *identity type*. Its terms can be thought of as paths identifying x and y. Given two such terms $p, q : x = y$, we can form another identity type

$$p = q,$$

whose terms can be thought of as homotopies between these paths. This process continues indefinitely.

A type therefore already carries higher-dimensional structure: terms, identifications between terms, identifications between identifications, and so on. These identifications behave like paths and higher homotopies in a space, so types naturally behave as ∞-groupoids. The higher structure is built into our foundations.

The structure identity principle. When studying mathematical structures in classical foundations, one must distinguish between properties that are invariant under isomorphism and those that depend on the chosen presentation. For example, given a group G, the property "G contains an element of order 2" is invariant under group isomorphism, whereas "the identity element of G is the number 0" is not. Indeed, the groups $(\{0, 1\}, +\text{mod } 2)$ and $(\{1, -1\}, \cdot)$ are isomorphic, but the latter property holds only for the first. Both properties can be expressed in set-theoretic foundations, but only the first describes the abstract group structure.

An inbuild feature of equality, both in set-theoretic foundations and in homotopy type theory, is the *indiscernibility principle*: if $x = y$, then any property of x also holds for y, and any additional structure on x can be transported to y, and vice-versa.

Homotopy type theory allows for an additional feature that is inconceivable in classical foundations: namely the *structure identity principle*. It connects equality with the mathematical notion of having the same structure. For example, given groups G and H, the terms of the identity type $G = H$ correspond precisely to group isomorphisms from G to H. Thus, giving an identification of two groups amounts to giving an isomorphism between them. The identity type records not just whether the groups can be identified, but also how they are identified.

Combined with indiscernibility, this implies that every property or construction of groups expressible in this framework is invariant under isomorphism. Invariance under isomorphism is therefore built into the foundations.

Synthetic category theory. The second aspect is the synthetic viewpoint itself. Rather than beginning with a concrete construction of an ∞-category, we will specify axioms describing how ∞-categories and the relationships between them behave. Our emphasis will therefore shift from the question "What is an ∞-category?" to "How does the collection of ∞-categories behave?" This requires a sufficiently rich axiomatic setting, but allows us to develop the theory directly from its structural properties.

To do this, we extend homotopy type theory so that we can also talk about non-invertible morphisms. In this framework, ∞-categories are basic types, and we retain the core principles of homotopy type theory, such as the structure identity principle.

Course overview

We will begin with a brief introduction to ordinary category theory, which we will call *strict category theory* throughout these notes. The aim is to become familiar with examples and basic constructions before encountering their ∞-categorical counterparts. We will then survey the foundations of homotopy type theory, following [Rij25, Uni13]. Next, we will pass to an extension of this type theory, which is inspired from [Riv25] and formulate the axioms of synthetic category theory in this extended type theory. This approach to the foundations

differs from the one presented in [CCNW21]. We will then develop category theory in this synthetic setting, following [CCNW21].

We plan to cover adjunctions, localisations, left and cocartesian fibrations, the Yoneda lemma, left and right Kan extensions, and limits and colimits. These names need not be familiar at this stage: the course will introduce them progressively, explaining both their meaning and their role.

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I. Strict category theory

In this first part, we introduce or recall various definitions and constructions in ordinary category theory that will later be generalized to the ∞-setting. Our aim is to build intuition for the kinds of structures and results we will encounter, so that the abstract setting introduced later feels less intimidating. We will also highlight some of the limitations of strict categories and explain why they can be too rigid to do category theory. For instance, working with *sets* of objects and morphisms encourages us to formulate notions in terms of equality of objects, which can lead to constructions that are not invariant under equivalences of strict categories. Understanding this distinction, and the difficulties it creates, will motivate the synthetic approach to category theory developed later in the course.

The topics covered in this part have been chosen specifically with the axioms of synthetic category theory in mind.

1. First definitions

We start by introducing strict categories, functors, and natural transformations, together with some examples. Throughout the course, we refer to ordinary categories as *strict categories*, reserving the term *categories* for their ∞-counterparts, namely synthetic categories, which will play a much more central role in this course.

1.1. Size issues

Russell's paradox tells us that the collection of all sets is not a set. To handle this issue, we will work with three nested universes of sets.

Definition 1.1. A **set-universe** is a set $\mathcal{U}$ whose elements are sets satisfying the following properties:

- (Infinite) the set of natural numbers $\mathbb{N}$ belongs to $\mathcal{U}$.
- (Subset closure) if $A \in \mathcal{U}$ and $B \subseteq A$, then $B \in \mathcal{U}$;
- (Power set closure) if $A \in \mathcal{U}$, then its power set $\mathcal{P}(A) \in \mathcal{U}$;
- (Union closure) if $I \in \mathcal{U}$ and $A_i \in \mathcal{U}$, for each $i \in I$, then $\bigcup_{i \in I} A_i \in \mathcal{U}$.

Remark 1.2. A similar proof to Russell's paradox applies here to show that $\mathcal{U} \notin \mathcal{U}$. By contradiction, suppose that $\mathcal{U} \in \mathcal{U}$. Then we can form the subset of $\mathcal{U}$

$$R \coloneqq \{A \in \mathcal{U} \mid A \notin A\} \subseteq \mathcal{U}$$

and, by hypothesis, we have that $R \in \mathcal{U}$. But this gives a contradiction since we have that $R \in R$ if and only if $R \notin R$.

Convention 1.3. We will assume that there exist set-universes

$$\mathcal{U}_{\text{small}} \subseteq \mathcal{U}_{\text{large}} \subseteq \mathcal{U}_{\text{huge}}$$

such that $\mathcal{U}_{\text{small}} \in \mathcal{U}_{\text{large}}$ and $\mathcal{U}_{\text{large}} \in \mathcal{U}_{\text{huge}}$.

Definition 1.4. We introduce the following terminology.

- A **set** or a **small set** is an element of $\mathcal{U}_{\text{small}}$.
- A **large set** is an element of $\mathcal{U}_{\text{large}}$.
- A **huge set** is an element of $\mathcal{U}_{\text{huge}}$.

Remark 1.5. Our definitions, statements, and constructions are relative to a chosen universe. We will sometimes apply these at a larger universe level, regarding "large" as the new "small" and "huge" as the new "large".

1.2. Strict categories

Definition 1.6. A **strict category** $\mathcal{C}$ consists of

- a large set $\mathrm{Ob}(\mathcal{C})$ of **objects** $x, y, z, \ldots$;
- for all objects $x, y \in \mathcal{C}$, a large set $\mathcal{C}(x, y)$ of **morphisms** $f\colon x \to y$ from x to y;
- for each object $x \in \mathcal{C}$, an **identity morphism** $\mathrm{id}_x\colon x \to x$;
- for all morphisms $f\colon x \to y$ and $g\colon y \to z$ in $\mathcal{C}$, a **composite morphism** $g \circ f\colon x \to z$; we also write simply gf for the composite;

satisfying the following axioms:

- composition is unital: for each morphism $f\colon x \to y$,
$$f \circ \mathrm{id}_x = f \quad \text{and} \quad \mathrm{id}_y \circ f = f;$$
- composition is associative: for all morphisms $f\colon x \to y$, $g\colon y \to z$, and $h\colon z \to t$ in $\mathcal{C}$,
$$(h \circ g) \circ f = h \circ (g \circ f).$$

Remark 1.7. By convention, a strict category has a large set of objects, which we refer to simply as its **object set**, and large sets of morphisms, which we refer to simply as its **hom sets**.

We will also consider **huge strict categories** with huge sets of objects and of morphisms.

Definition 1.8. A strict category $\mathcal{C}$ is

- **locally small** if, for all objects $x, y \in \mathcal{C}$, the hom set $\mathcal{C}(x, y)$ is small;
- **small** if its object set is small and it is locally small.

Example 1.9. We have the following examples of strict categories.

- The strict category $\mathcal{S}et$ of sets and functions.
- The strict category $\mathcal{S}et_\bullet$ of pointed sets and basepoint-preserving functions.
- For a field k, the strict category $\mathcal{V}ect_k$ of k-vector spaces and k-linear maps.
- The strict category $\mathcal{T}op$ of topological spaces and continuous functions.

All these strict categories are locally small, but not small.

Definition 1.10. Let $\mathcal{C}$ be a strict category. A morphism $f\colon x \to y$ in $\mathcal{C}$ is an **isomorphism** if there is a morphism $g\colon y \to x$ in $\mathcal{C}$ such that $g \circ f = \mathrm{id}_x$ and $f \circ g = \mathrm{id}_y$. We write $f\colon x \xrightarrow{\cong} y$ or $f\colon x \cong y$. We call g an **inverse** of f.

We say that two objects x and y in $\mathcal{C}$ are **isomorphic** if there is an isomorphism $f\colon x \xrightarrow{\cong} y$ in $\mathcal{C}$. We write $x \cong y$.

Example 1.11. The strict categories in Example 1.9 have the following isomorphisms.

- An isomorphism in $\mathcal{S}et$ is a bijection.
- An isomorphism in $\mathcal{S}et_\bullet$ is a basepoint-preserving bijection.
- An isomorphism in $\mathcal{V}ect_k$ is a linear bijection.
- An isomorphism in $\mathcal{T}op$ is a homeomorphism.

Proposition 1.12. *Let $\mathcal{C}$ be a strict category. An isomorphism $f\colon x \to y$ in $\mathcal{C}$ has a unique inverse, which we denote by $f^{-1}\colon y \to x$.*

Proof. Suppose that $g, h\colon y \to x$ are two inverses of f in $\mathcal{C}$. Then

$$g = g \circ \mathrm{id}_y = g \circ (f \circ h) = (g \circ f) \circ h = \mathrm{id}_x \circ h = h. \qquad \square$$

1.3. **Functors between strict categories**

Definition 1.13. Given strict categories $\mathcal{C}$ and $\mathcal{D}$, a **functor** $F\colon \mathcal{C} \to \mathcal{D}$ consists of:

- a function $\mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D})$ sending an object $x \in \mathcal{C}$ to an object $Fx \in \mathcal{D}$;
- for all objects $x, y \in \mathcal{C}$, a function $\mathcal{C}(x, y) \to \mathcal{D}(Fx, Fy)$ sending a morphism $f\colon x \to y$ in $\mathcal{C}$ to a morphism $Ff\colon Fx \to Fy$ in $\mathcal{D}$;

satisfying the following axioms:

- F preserves identities: for each object $x \in \mathcal{C}$,
$$F(\mathrm{id}_x) = \mathrm{id}_{Fx};$$
- F preserves composites: for all morphisms $f\colon x \to y$ and $g\colon y \to z$ in $\mathcal{C}$,
$$F(g \circ f) = Fg \circ Ff.$$

Example 1.14. We have the following examples of functors.

- The forgetful functors $U\colon \mathcal{S}et_\bullet \to \mathcal{S}et$, $U\colon \mathcal{V}ect_k \to \mathcal{S}et$ and $U\colon \mathcal{T}op \to \mathcal{S}et$ sending a pointed set, a k-vector space, or a topological space to its underlying set.
- The functor $(-)_+\colon \mathcal{S}et \to \mathcal{S}et_\bullet$ which freely adjoins a basepoint to a set S, namely $S_+ \coloneqq (S \amalg \{*\}, *)$.
- The functor $\bigoplus_{(-)} k\colon \mathcal{S}et \to \mathcal{V}ect_k$ which sends a set S to the k-vector space with basis S, namely $\bigoplus_S k$.
- The discrete topology functor $D\colon \mathcal{S}et \to \mathcal{T}op$ equipping a set S with the discrete topology, namely $(S, \mathcal{P}(S))$, where $\mathcal{P}(S)$ is the set of subsets of S.
- The indiscrete topology functor $I\colon \mathcal{S}et \to \mathcal{T}op$ equipping a set S with the indiscrete topology, namely $(S, \{\emptyset, S\})$.

Proposition 1.15. *Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor between strict categories. Then F preserves isomorphisms, i.e., if f is an isomorphism in $\mathcal{C}$, then Ff is an isomorphism in $\mathcal{D}$.*

Proof. Exercise. $\square$

Construction 1.16. There is a strict category $\mathcal{C}at_s$ with

- object set $\mathrm{Ob}(\mathcal{C}at_s)$ consisting of the small strict categories;
- for all objects $\mathcal{C}, \mathcal{D} \in \mathcal{C}at_s$, hom set $\mathcal{C}at_s(\mathcal{C}, \mathcal{D})$ consisting of the functors $\mathcal{C} \to \mathcal{D}$;
- for each object $\mathcal{C} \in \mathcal{C}at_s$, **identity functor** $\mathrm{id}_\mathcal{C}\colon \mathcal{C} \to \mathcal{C}$ acting as the identity on objects and on morphisms;
- for all morphisms $F\colon \mathcal{C} \to \mathcal{D}$ and $G\colon \mathcal{D} \to \mathcal{E}$ in $\mathcal{C}at_s$, **composite functor** $GF\colon \mathcal{C} \to \mathcal{E}$ sending
  - an object $x \in \mathcal{C}$ to the object $G(F(x)) \in \mathcal{E}$;

– a morphism $f\colon x \to y$ in $\mathcal{C}$ to the morphism $G(F(f))\colon G(F(x)) \to G(F(y))$ in $\mathcal{E}$.

This is a well-defined strict category (Exercise).

Similarly, there is a huge strict category $\mathcal{CAT}_s$ of (large) strict categories and functors.

Remark 1.17. The strict category $\mathcal{C}at_s$ of small strict categories and functors defines an object of $\mathcal{CAT}_s$. As for the other examples, the strict category $\mathcal{C}at_s$ is locally small, but not small.

Construction 1.18. Given a strict category $\mathcal{C}$, there is a strict category $\mathcal{C}^{\mathrm{op}}$, called the **opposite strict category** of $\mathcal{C}$, with

- object set $\mathrm{Ob}(\mathcal{C})$;
- for all objects $x, y \in \mathcal{C}$, hom set
$$\mathcal{C}^{\mathrm{op}}(x, y) \coloneqq \mathcal{C}(y, x);$$
given a morphism $f\colon y \to x$ in $\mathcal{C}$, we write $f^{\mathrm{op}}\colon x \to y$ for the corresponding morphism in $\mathcal{C}^{\mathrm{op}}$;
- for each object $x \in \mathcal{C}$, identity morphism $(\mathrm{id}_x)^{\mathrm{op}}$;
- for all morphisms $f^{\mathrm{op}}\colon x \to y$ and $g^{\mathrm{op}}\colon y \to z$ in $\mathcal{C}^{\mathrm{op}}$, composite morphism
$$g^{\mathrm{op}} \circ f^{\mathrm{op}} \coloneqq (f \circ g)^{\mathrm{op}}\colon x \to z.$$

This is a well-defined strict category (Exercise).

Remark 1.19. A **contravariant functor** from $\mathcal{C}$ to $\mathcal{D}$ is a functor $F\colon \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$. It consists of

- a function $\mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D})$ sending an object $x \in \mathcal{C}$ to an object $Fx \in \mathcal{D}$;
- for all objects $x, y \in \mathcal{C}$, a function $\mathcal{C}(x, y) \to \mathcal{D}(Fy, Fx)$ sending a morphism $f\colon x \to y$ in $\mathcal{C}$ to a morphism $Ff\colon Fy \to Fx$ in $\mathcal{D}$;

satisfying the following axioms:

- F preserves identities;
- F preserves compositions: for all morphisms $f\colon x \to y$ and $g\colon y \to z$ in $\mathcal{C}$,
$$F(g \circ f) = Ff \circ Fg.$$

Construction 1.20. Let $\mathcal{C}$ be a locally small strict category and $c \in \mathcal{C}$ be an object. There is a functor
$$\mathcal{C}(-, c)\colon \mathcal{C}^{\mathrm{op}} \to \mathcal{S}et,$$
called the **representable functor** at c, sending:

- an object $x \in \mathcal{C}$ to the (small) set of morphisms $\mathcal{C}(x, c)$;
- a morphism $f\colon x \to y$ in $\mathcal{C}$ to the function
$$f^*\colon \mathcal{C}(y, c) \to \mathcal{C}(x, c), \quad (g\colon y \to c) \mapsto (gf\colon x \to c).$$

This is a well-defined functor (Exercise).

1.4. Natural transformations between functors

Definition 1.21. Given functors $F, G\colon \mathcal{C} \to \mathcal{D}$ between strict categories, a **natural transformation** $\alpha\colon F \Rightarrow G$ consists of:

- for each object $x \in \mathcal{C}$, a morphism $\alpha_x\colon Fx \to Gx$ in $\mathcal{D}$, called the **component** of α at x;

satisfying the following axiom:

- α is natural: for each morphism $f\colon x \to y$ in $\mathcal{C}$, the following square commutes

$$\begin{array}{ccc} Fx & \xrightarrow{\alpha_x} & Gx \\ {\scriptstyle Ff}\downarrow & & \downarrow{\scriptstyle Gf} \\ Fy & \xrightarrow[\alpha_y]{} & Gy \end{array}$$

i.e., $Gf \circ \alpha_x = \alpha_y \circ Ff$.

Construction 1.22. Let $\mathcal{C}$ be a locally small strict category and $f\colon c \to d$ be a morphism in $\mathcal{C}$. There is a natural transformation

$$\mathcal{C}(-, f)\colon \mathcal{C}(-, c) \Rightarrow \mathcal{C}(-, d)$$

whose component at an object $x \in \mathcal{C}$ is the function

$$f_*\colon \mathcal{C}(x, c) \to \mathcal{C}(x, d), \quad (g\colon x \to c) \mapsto (fg\colon x \to d).$$

This is well-defined (Exercise).

Definition 1.23. Given functors $F, G\colon \mathcal{C} \to \mathcal{D}$ between strict categories, a natural transformation $\alpha\colon F \Rightarrow G$ is a **natural isomorphism** if, for each object $x \in \mathcal{C}$, its component $\alpha_x\colon Fx \to Gx$ is an isomorphism in $\mathcal{D}$.

We sometimes say that, for all objects $x \in \mathcal{C}$, there is an isomorphism $Fx \cong Gx$ which is **natural** in x.

Construction 1.24. Given strict categories $\mathcal{C}$ and $\mathcal{D}$, there is a strict category $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$, called the **functor strict category**, with

- object set $\mathrm{Ob}(\mathrm{Fun}(\mathcal{C}, \mathcal{D})) \coloneqq \mathcal{CAT}_s(\mathcal{C}, \mathcal{D})$ consisting of the functors $\mathcal{C} \to \mathcal{D}$;
- for all objects $F, G \in \mathrm{Fun}(\mathcal{C}, \mathcal{D})$, hom set $\mathrm{Fun}(\mathcal{C}, \mathcal{D})(F, G)$ consisting of the natural transformations $F \Rightarrow G$;
- for each object $F \in \mathrm{Fun}(\mathcal{C}, \mathcal{D})$, **identity natural transformation** $\mathrm{id}_F\colon F \Rightarrow F$ having component at an object $c \in \mathcal{C}$ given by
$$(\mathrm{id}_F)_c \coloneqq \mathrm{id}_{Fc}\colon Fc \to Fc;$$
- for all morphisms $\alpha\colon F \Rightarrow G$ and $\beta\colon G \Rightarrow H$ in $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$, **composite natural transformation** $\beta \circ \alpha\colon F \Rightarrow H$ having component at an object $c \in \mathcal{C}$ given by the composite
$$(\beta \circ \alpha)_c \coloneqq \beta_c \circ \alpha_c\colon Fc \to Hc.$$

This is a well-defined strict category (Exercise).

Proposition 1.25. *Given strict categories $\mathcal{C}$ and $\mathcal{D}$, a natural transformation is an isomorphism in* $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ *if and only if it is a natural isomorphism.*

Proof. Exercise. □

Construction 1.26. The **terminal strict category** is the strict category $[0]$ with a single object $*$ and a single morphism id_*.

Proposition 1.27. *Given a strict category $\mathcal{C}$, there is an isomorphism of strict categories*

$$\mathrm{Fun}([0], \mathcal{C}) \cong \mathcal{C}$$

which is natural in $\mathcal{C}$.

Proof. Exercise. □

Remark 1.28. As a consequence of this result, we have that an object $c \in \mathcal{C}$ corresponds to a unique functor $c\colon [0] \to \mathcal{C}$, which we also call c.

Construction 1.29. Given a functor $F\colon \mathcal{C} \to \mathcal{D}$ between strict categories and a strict category $\mathcal{E}$, there is a functor

$$F^*\colon \mathrm{Fun}(\mathcal{D},\mathcal{E}) \to \mathrm{Fun}(\mathcal{C},\mathcal{E}),$$

called the **precomposition functor** with F, sending

- a functor $G\colon \mathcal{D} \to \mathcal{E}$ to the composite $GF\colon \mathcal{C} \to \mathcal{E}$;
- a natural transformation $\alpha\colon G \Rightarrow H$ between functors $G, H\colon \mathcal{D} \to \mathcal{E}$ to the natural transformation $\alpha_F\colon GF \Rightarrow HF$ whose component at an object $x \in \mathcal{C}$ is the morphism in $\mathcal{E}$

$$\alpha_{Fx}\colon G(Fx) \to H(Fx).$$

There is also a functor

$$F_*\colon \mathrm{Fun}(\mathcal{E},\mathcal{C}) \to \mathrm{Fun}(\mathcal{E},\mathcal{D}),$$

called the **postcomposition functor** with F, sending

- a functor $G\colon \mathcal{E} \to \mathcal{C}$ to the composite $FG\colon \mathcal{E} \to \mathcal{D}$;
- a natural transformation $\alpha\colon G \Rightarrow H$ between functors $G, H\colon \mathcal{E} \to \mathcal{C}$ to the natural transformation $F\alpha\colon FG \Rightarrow FH$ whose component at an object $x \in \mathcal{E}$ is the morphism in $\mathcal{D}$

$$F(\alpha_x)\colon F(Gx) \to F(Hx).$$

These are well-defined functors (Exercise).

2. Yoneda lemma

We now turn to one of the fundamental results of category theory, the *Yoneda lemma*. An important consequence of this result is that an object in a strict category is determined, up to isomorphism, by its representable functor. Hence we can understand an object (up to isomorphism) through its relationships with other objects. It also highlights the central role of the strict category $\mathcal{S}et$ in strict category theory. For instance, we can detect isomorphisms in a strict category by proving that certain functions between sets are bijections.

Theorem 2.1 (Yoneda lemma)**.** *Let $\mathcal{C}$ be a locally small strict category. For all objects $c \in \mathcal{C}$ and functors $F\colon \mathcal{C}^{\mathrm{op}} \to \mathcal{S}et$, there is a bijection*

$$\mathrm{Fun}(\mathcal{C}^{\mathrm{op}},\mathcal{S}et)(\mathcal{C}(-,c),F) \cong Fc$$

which is natural in both c and F.

Proof. Evaluation at the identity morphism defines a function

$$\mathrm{Fun}(\mathcal{C}^{\mathrm{op}},\mathcal{S}et)(\mathcal{C}(-,c),F) \to Fc, \quad (\alpha\colon \mathcal{C}(-,c) \Rightarrow F) \mapsto \alpha_c(\mathrm{id}_c),$$

where $\alpha_c\colon \mathcal{C}(c,c) \to Fc$ is the function given by the component of α at c.

Next, consider the function

$$Fc \to \mathrm{Fun}(\mathcal{C}^{\mathrm{op}},\mathcal{S}et)(\mathcal{C}(-,c),F), \quad u \mapsto \alpha_u$$

sending an element $u \in Fc$ to $\alpha_u\colon \mathcal{C}(-,c) \Rightarrow F$ with component at an object $x \in \mathcal{C}$ given by the function

$$(\alpha_u)_x\colon \mathcal{C}(x,c) \to Fx, \quad (f\colon x \to c) \mapsto Ff(u),$$

where $Ff\colon Fc \to Fx$ is the function given by the image of f under F. This is a well-defined natural transformation (Exercise).

These functions are mutually inverse and natural in c and F (Exercise). □

Construction 2.2. Let $\mathcal{C}$ be a locally small strict category. There is a functor

$$\mathbf{y}_{\mathcal{C}} \colon \mathcal{C} \to \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}et)$$

called the **Yoneda functor**, sending

- an object $c \in \mathcal{C}$ to the representable functor $\mathcal{C}(-, c) \colon \mathcal{C}^{\mathrm{op}} \to \mathcal{S}et$;
- a morphism $f \colon c \to d$ in $\mathcal{C}$ to the natural transformation $\mathcal{C}(-, f) \colon \mathcal{C}(-, c) \Rightarrow \mathcal{C}(-, d)$.

This is well-defined (Exercise).

Definition 2.3. A functor $F \colon \mathcal{C} \to \mathcal{D}$ between strict categories is **fully faithful** if, for all objects $x, y \in \mathcal{C}$, the induced function

$$\mathcal{C}(x, y) \to \mathcal{D}(Fx, Fy)$$

is a bijection.

Corollary 2.4. *Let $\mathcal{C}$ be a locally small strict category. The Yoneda functor*

$$\mathbf{y}_{\mathcal{C}} \colon \mathcal{C} \to \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}et)$$

is fully faithful.

Proof. Exercise. □

Proposition 2.5. *Let $F \colon \mathcal{C} \to \mathcal{D}$ be a fully faithful functor. Then F reflects isomorphisms, i.e., if f is a morphism in $\mathcal{C}$ such that Ff is an isomorphism in $\mathcal{D}$, then f is an isomorphism in $\mathcal{C}$.*

Moreover, two objects $x, y \in \mathcal{C}$ are isomorphic if and only if their images $Fx, Fy \in \mathcal{D}$ are isomorphic.

Proof. Exercise. □

Corollary 2.6. *Let $\mathcal{C}$ be a locally small strict category. The following are equivalent for a morphism $f \colon c \to d$ in $\mathcal{C}$.*

(i) The morphism f is an isomorphism in $\mathcal{C}$.

(ii) For all objects $x \in \mathcal{C}$, the induced function

$$f_* \colon \mathcal{C}(x, c) \to \mathcal{C}(x, d)$$

is a bijection.

(iii) For all objects $x \in \mathcal{C}$, the induced function

$$f^* \colon \mathcal{C}(d, x) \to \mathcal{C}(c, x)$$

is a bijection.

Proof. Since the Yoneda functor $\mathbf{y}_{\mathcal{C}} \colon \mathcal{C} \to \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}et)$ is fully faithful, it preserves and reflects isomorphisms. Hence the morphism $f \colon c \to d$ is an isomorphism in $\mathcal{C}$ if and only if the natural transformation $\mathcal{C}(-, f) \colon \mathcal{C}(-, c) \Rightarrow \mathcal{C}(-, d)$ is an isomorphism in $\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S}et)$. By Proposition 1.25, this holds if and only if it is a natural isomorphism, i.e., for every object $x \in \mathcal{C}$, the induced function

$$f_* \colon \mathcal{C}(x, c) \to \mathcal{C}(x, d)$$

is a bijection. This proves that (i) and (ii) are equivalent.

Next, observe that f is an isomorphism in $\mathcal{C}$ if and only if $f^{\mathrm{op}} \colon d \to c$ is an isomorphism in $\mathcal{C}^{\mathrm{op}}$ (Exercise). Then the equivalence between (i) and (ii) for $\mathcal{C}^{\mathrm{op}}$ implies that f^{op} is an isomorphism in $\mathcal{C}^{\mathrm{op}}$ if and only if, for every object $x \in \mathcal{C}$, the induced function

$$f^{\mathrm{op}}_* = f^* \colon \mathcal{C}^{\mathrm{op}}(x, d) = \mathcal{C}(d, x) \to \mathcal{C}(c, x) = \mathcal{C}^{\mathrm{op}}(x, c)$$

is a bijection. This proves that (i) and (iii) are equivalent. □

Corollary 2.7. *Let $\mathcal{C}$ be a locally small strict category. Two objects $x, y \in \mathcal{C}$ are isomorphic if and only if their representable functors $\mathcal{C}(-, x)$ and $\mathcal{C}(-, y)$ are naturally isomorphic.*

Proof. This follows from the preceding proposition applied to the fully faithful Yoneda functor, together with Proposition 1.25. □

3. Adjunctions and equivalences

We now turn to two important concepts in strict category theory, namely *adjunctions* and *equivalences*. Adjunctions provide a way to describe universal constructions and organize them into functors. Equivalences express when two strict categories should be thought of as being the "same".

From a categorical point of view, two isomorphic objects should be interchangeable, even if they are not equal. However, an isomorphism of strict categories requires a bijection between the sets of objects, making it too restrictive for this purpose. Equivalences provide a weakening of the notion of isomorphism between strict categories and capture the intended notion of sameness: they detect objects only up to isomorphism, rather than up to equality.

A fundamental difficulty with the set-theoretic description of strict categories is that equality and isomorphism of objects remain distinct notions. Consequently, properties and constructions can depend on literal equalities between objects and fail to be well-behaved with respect to isomorphisms. We will illustrate this with examples of properties and constructions that are not invariant under replacing objects by isomorphic ones or strict categories by equivalent ones. Further examples will appear in the following sections.

The synthetic approach will address this foundational issue by not allowing a distinction between equality and isomorphism of objects, so that invariance under isomorphism is built into the theory. Similarly, the appropriate notion of isomorphism between synthetic categories will generalize that of an equivalence between strict categories.

3.1. Adjunctions

Definition 3.1. Given strict categories $\mathcal{C}$ and $\mathcal{D}$, an **adjunction** between $\mathcal{C}$ and $\mathcal{D}$ consists of:

- a functor $L\colon \mathcal{C} \to \mathcal{D}$, called the **left adjoint**;
- a functor $R\colon \mathcal{D} \to \mathcal{C}$, called the **right adjoint**;
- a natural transformation $\eta\colon \mathrm{id}_{\mathcal{C}} \Rightarrow RL$, called the **unit**;
- a natural transformation $\varepsilon\colon LR \Rightarrow \mathrm{id}_{\mathcal{D}}$, called the **counit**;

such that, for all objects $x \in \mathcal{C}$ and $y \in \mathcal{D}$, the following triangles commute in $\mathcal{D}$ and $\mathcal{C}$, respectively:



We write $L\colon \mathcal{C} \rightleftarrows \mathcal{D} : R$ or $L \dashv R$ for such an adjunction.

Example 3.2. We have the following examples of adjunctions between strict categories.

- The free-forgetful adjunctions

$$(-)_+\colon \mathcal{S}et \rightleftarrows \mathcal{S}et_\bullet : U \quad \text{and} \quad \bigoplus_{(-)} k\colon \mathcal{S}et \rightleftarrows \mathcal{V}ect_k : U.$$

- The discrete-forgetful and forgetful-indiscrete adjunctions

$$D\colon \mathcal{S}et \rightleftarrows \mathcal{T}op : U \quad \text{and} \quad U\colon \mathcal{T}op \rightleftarrows \mathcal{S}et : I.$$

See Example 1.14 for the definition of the functors involved.

Theorem 3.3 (Adjunctions representably)**.** *Given functors $L\colon \mathcal{C} \to \mathcal{D}$ and $R\colon \mathcal{D} \to \mathcal{C}$ between strict categories, there is a unit and a counit making the pair (L, R) into an adjunction $L \dashv R$ if and only if, for all objects $x \in \mathcal{C}$ and $y \in \mathcal{D}$, there is a bijection*

$$\mathcal{D}(Lx, y) \cong \mathcal{C}(x, Ry)$$

which is natural in x and y.

Proof. Given an adjunction $L \dashv R$, the unit $\eta\colon \mathrm{id}_{\mathcal{C}} \Rightarrow RL$ induces a function

$$\mathcal{D}(Lx, y) \xrightarrow{R} \mathcal{C}(RLx, Ry) \xrightarrow{(\eta_x)^*} \mathcal{C}(x, Ry), \quad (g\colon Lx \to y) \mapsto R(g) \circ \eta_x,$$

and the counit $\varepsilon\colon LR \Rightarrow \mathrm{id}_{\mathcal{D}}$ induces a function

$$\mathcal{C}(x, Ry) \xrightarrow{L} \mathcal{D}(Lx, LRy) \xrightarrow{(\varepsilon_y)_*} \mathcal{D}(Lx, y), \quad (f\colon x \to Ry) \mapsto \varepsilon_y \circ L(f).$$

These are mutually inverse and natural in x and y (Exercise).

Conversely, suppose that, for all objects $x \in \mathcal{C}$ and $y \in \mathcal{D}$, we are given natural bijections

$$\varphi_{x,y}\colon \mathcal{D}(Lx, y) \cong \mathcal{C}(x, Ry).$$

Define $\eta\colon \mathrm{id}_{\mathcal{C}} \Rightarrow RL$ whose component at an object $x \in \mathcal{C}$ is given by

$$\eta_x \coloneqq \varphi_{x,Lx}(\mathrm{id}_{Lx})\colon x \to RLx.$$

Define $\varepsilon\colon LR \Rightarrow \mathrm{id}_{\mathcal{D}}$ whose component at an object $y \in \mathcal{D}$ is given by the morphism

$$\varepsilon_y \coloneqq \varphi^{-1}_{Ry,y}(\mathrm{id}_{Ry})\colon LRy \to y.$$

These are well-defined natural transformations satisfying the triangle identities (Exercise). This proves that $L \dashv R$ is an adjunction. □

3.2. **Equivalences of strict categories**

Recall that a functor $F\colon \mathcal{C} \to \mathcal{D}$ between strict categories is an isomorphism in $\mathcal{CAT}_s$ if there is a functor $G\colon \mathcal{D} \to \mathcal{C}$ such that $GF = \mathrm{id}_{\mathcal{C}}$ and $FG = \mathrm{id}_{\mathcal{D}}$.

Since we have a notion of morphisms between functors, namely the natural transformations, we can weaken the notion of sameness between strict categories as follows.

Definition 3.4. A functor $F\colon \mathcal{C} \to \mathcal{D}$ between strict categories is an **equivalence** if there is a functor $G\colon \mathcal{D} \to \mathcal{C}$ and natural isomorphisms $\eta\colon \mathrm{id}_{\mathcal{C}} \overset{\cong}{\Longrightarrow} GF$ and $\varepsilon\colon FG \overset{\cong}{\Longrightarrow} \mathrm{id}_{\mathcal{D}}$.

Example 3.5. The following classes of functors give special cases of equivalences.

- Any isomorphism in $\mathcal{CAT}_s$ is an equivalence of strict categories.
- Given an adjunction $L \dashv R$ such that the unit and the counit are natural isomorphisms, both L and R are equivalences of strict categories; we refer to this as an **adjoint equivalence**.

Fact: every equivalence of strict categories can be promoted to an adjoint equivalence; see [Rie16, Corollary 4.3.6]. We won't use this fact in this course.

Definition 3.6. A functor $F\colon \mathcal{C} \to \mathcal{D}$ between strict categories is **essentially surjective** if, for all objects $d \in \mathcal{D}$, there is an object $c \in \mathcal{C}$ and an isomorphism $\varphi\colon Fc \xrightarrow{\cong} d$ in $\mathcal{D}$.

Theorem 3.7 (Fundamental Theorem)**.** *A functor $F\colon \mathcal{C} \to \mathcal{D}$ between strict categories is an equivalence if and only if it is essentially surjective on objects and fully faithful.*

Proof. Suppose that F is an equivalence so that we have a functor $G\colon \mathcal{D} \to \mathcal{C}$ and natural isomorphisms $\eta\colon \mathrm{id}_{\mathcal{C}} \overset{\cong}{\Longrightarrow} GF$ and $\varepsilon\colon FG \overset{\cong}{\Longrightarrow} \mathrm{id}_{\mathcal{D}}$. Given an object $d \in \mathcal{D}$, the object $Gd \in \mathcal{C}$ and the isomorphism in $\mathcal{D}$

$$\varepsilon_d\colon FGd \xrightarrow{\cong} d$$

provide the desired data for F to be essentially surjective. Next, given objects $x, y \in \mathcal{C}$, we show that the induced function

$$\mathcal{C}(x,y) \to \mathcal{D}(Fx, Fy)$$

is a bijection. Its inverse is given by the function

$$\mathcal{D}(Fx,Fy) \to \mathcal{C}(x,y), \quad (g\colon Fx \to Fy) \mapsto (x \xrightarrow{\eta_x} GFx \xrightarrow{Gg} GFy \xrightarrow{\eta_y^{-1}} y).$$

Naturality of $\eta\colon \mathrm{id}_{\mathcal{C}} \Rightarrow GF$ and $\varepsilon\colon FG \Rightarrow \mathrm{id}_{\mathcal{D}}$ implies that these functions are mutual inverses (Exercise). This proves that F is fully faithful.

Conversely, suppose that F is essentially surjective on objects and fully faithful. We construct a functor $G\colon \mathcal{D} \to \mathcal{C}$ and a natural isomorphism $\varepsilon\colon FG \overset{\cong}{\Longrightarrow} \mathrm{id}_{\mathcal{D}}$ simultaneously. Let $d \in \mathcal{D}$ be an object. Since F is essentially surjective, there is an object $c \in \mathcal{C}$ and an isomorphism $\varphi\colon Fc \xrightarrow{\cong} d$. We set $Gd \coloneqq c$ and $\varepsilon_d \coloneqq \varphi\colon FGd \xrightarrow{\cong} d$. Next, let $g\colon x \to y$ be a morphism in $\mathcal{D}$ and consider the composite in $\mathcal{D}$

$$FGx \xrightarrow{\varepsilon_x} x \xrightarrow{g} y \xrightarrow{\varepsilon_y^{-1}} FGy.$$

Since F is fully faithful, there is a unique morphism $f\colon Gx \to Gy$ in $\mathcal{C}$ with $Ff = \varepsilon_y^{-1} \circ g \circ \varepsilon_x$. We set $Gg \coloneqq f$. Then G is a well-defined functor and the isomorphisms ε_d assemble into a natural transformation (Exercise).

Finally, we construct the natural isomorphism $\eta\colon \mathrm{id}_{\mathcal{C}} \overset{\cong}{\Longrightarrow} GF$. Let $c \in \mathcal{C}$ be an object. Consider the morphism in $\mathcal{D}$

$$\varepsilon_{Fc}^{-1}\colon Fc \xrightarrow{\cong} FGFc.$$

Since F is fully faithful, there is a unique isomorphism $f\colon c \xrightarrow{\cong} GFc$ in $\mathcal{C}$ with $Ff = \varepsilon_{Fc}^{-1}$. We set $\eta_c \coloneqq f$. Then the isomorphisms η_c assemble into a natural transformation (Exercise). □

Construction 3.8. The **walking isomorphism** is the strict category $\mathcal{I} \coloneqq \{0 \cong 1\}$ with two objects 0 and 1, whose only morphisms are the identities and mutually inverse isomorphisms $f\colon 0 \to 1$ and $f^{-1}\colon 1 \to 0$. Observe that the hom sets of $\mathcal{I}$ are all singletons.

Proposition 3.9. *The functor $0\colon [0] \to \mathcal{I}$ is an equivalence of strict categories.*

Proof. It is surjective on objects up to isomorphisms, since $0 \cong 1$ in $\mathcal{I}$, and it is fully faithful since, for the unique object $* \in [0]$, we have a bijection

$$[0](*,*) = \{\mathrm{id}_*\} \xrightarrow{\cong} \{\mathrm{id}_0\} = \mathcal{I}(0,0).$$ □

3.3. **Stability under isomorphism and invariance under equivalence**

As mentioned in the section's introduction, we want to study objects in a strict category up to isomorphism and, accordingly, strict categories up to equivalence. Some notions that we can define at the point-set level may fail to be invariant under isomorphisms between objects or equivalences between strict categories and are therefore unsuitable as intrinsic notions of category theory. We call a notion *iso-invariant* if it is invariant under replacing an object by an isomorphic one, and *iso-sensitive* otherwise. Similarly, we call a notion *equiv-invariant* if it is invariant under replacing a strict category by an equivalent one, and *equiv-sensitive* otherwise. The following examples illustrate these distinctions.

Construction 3.10. The **walking morphism** is the strict category $[1]$ with two objects 0 and 1, whose only morphisms are the identities and a single morphism $0 \to 1$.

Example 3.11 (Iso-sensitive notion). Let $\mathcal{C}$ be a strict category containing a non-identity isomorphism $f\colon x \xrightarrow{\cong} y$. Consider the arrow strict category $\mathrm{Fun}([1],\mathcal{C})$, whose objects are morphisms in $\mathcal{C}$ and whose morphisms are commutative squares in $\mathcal{C}$. The property for an object of $\mathrm{Fun}([1],\mathcal{C})$ to be an identity morphism in $\mathcal{C}$ is not stable under isomorphisms in $\mathrm{Fun}([1],\mathcal{C})$. Indeed, we have a commutative square in $\mathcal{C}$

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ {\scriptstyle \mathrm{id}_x}\downarrow{\scriptstyle \cong} & & {\scriptstyle f^{-1}}\downarrow{\scriptstyle \cong} \\ x & \xrightarrow[\mathrm{id}_x]{} & x \end{array}$$

where both vertical morphisms are isomorphisms, which provides an isomorphism between f and id_x in $\mathrm{Fun}([1],\mathcal{C})$. However, the morphism f is not an identity morphism in $\mathcal{C}$.

In the next section, we will consider universal properties for objects, which will all be iso-invariant.

We now turn to an example of a property of a strict category that is equiv-sensitive.

Construction 3.12. Every large set S can be regarded as a strict category, also denoted by S, with object set S and only identity morphisms.

Example 3.13 (Equiv-sensitive notion). Consider the property for a strict category $\mathcal{C}$ to be a (large) set, i.e., all morphisms in $\mathcal{C}$ are identities. This property is not invariant under equivalence of strict categories. Indeed, the functor $0\colon [0] \to \mathcal{I}$ is an equivalence of strict categories. However, the strict category $[0]$ is clearly a set, whereas $\mathcal{I}$ is not, as it has non-identity morphisms f and f^{-1}.

We will see in the exercises that the correct replacement of the property of "being a (large) set" is that of a strict category whose hom sets are either empty or singletons and whose morphisms are all isomorphisms.

However, if we restrict the condition above to endomorphisms, i.e., morphisms of the form $x \to x$, then this property becomes equiv-invariant.

Example 3.14 (Equiv-invariant notion). Consider the property for a strict category $\mathcal{C}$ that all endomorphisms in $\mathcal{C}$ are identities. Unlike the one above, this property is invariant under equivalence. Indeed, if $F\colon \mathcal{C} \xrightarrow{\simeq} \mathcal{D}$ is an equivalence between strict categories, then, for every object $y \in \mathcal{D}$, there is an object $x \in \mathcal{C}$ together with an isomorphism $f\colon Fx \cong y$. Using that

F is fully faithful and conjugation by f, we obtain bijections

$$\mathcal{C}(x,x) \xrightarrow{\cong} \mathcal{D}(Fx,Fx) \xrightarrow{\cong} \mathcal{D}(y,y), \quad (g\colon x \to x) \mapsto (f \circ Fg \circ f^{-1}\colon y \to y).$$

If all endomorphisms in $\mathcal{C}$ are identities, then $\mathcal{C}(x,x) = \{\mathrm{id}_x\}$. Hence $\mathcal{D}(y,y)$ is also a singleton and therefore $\mathcal{D}(y,y) = \{\mathrm{id}_y\}$. So all endomorphisms in $\mathcal{D}$ are also identities.

4. Universal constructions in strict categories

We now turn to universal constructions in a strict category. Rather than introducing them in full generality, we will focus on initial and terminal objects, products, coproducts, and pullbacks. In the synthetic setting, their counterparts will be introduced directly as basic constructions of synthetic categories. These examples will help us become familiar with universal properties and learn how to work with them in general.

4.1. Terminal and initial objects

Definition 4.1. Let $\mathcal{C}$ be a strict category.

- An object $t \in \mathcal{C}$ is **terminal** if, for every object $x \in \mathcal{C}$, there is a unique morphism $!\colon x \to t$ in $\mathcal{C}$.
- An object $i \in \mathcal{C}$ is **initial** if, for every object $x \in \mathcal{C}$, there is a unique morphism $!\colon i \to x$ in $\mathcal{C}$.

Example 4.2. The strict categories in Example 1.9 have the following terminal and initial objects.

- The singleton set $*$ is a terminal object in $\mathcal{S}et$, and the empty set $\emptyset$ is an initial one.
- The singleton set $*$ is both a terminal and an initial object in $\mathcal{S}et_\bullet$.
- The zero k-vector space 0 is both a terminal and an initial object in $\mathcal{V}ect_k$.
- The singleton topological space $*$ is a terminal object in $\mathcal{T}op$, and the empty topological space $\emptyset$ is an initial one.

Proposition 4.3. *Let $\mathcal{C}$ be a strict category.*

- *(Uniqueness) Terminal objects are unique up to a unique isomorphism, i.e., if $t, t' \in \mathcal{C}$ are terminal objects, then there is a unique isomorphism $t \cong t'$ in $\mathcal{C}$.*
- *(Representability) An object $t \in \mathcal{C}$ is terminal if and only if, for every object $x \in \mathcal{C}$, there is a bijection $\mathcal{C}(x,t) \cong *$.*
- *(Iso-invariance) If $t \in \mathcal{C}$ is a terminal object and $t \cong t'$ in $\mathcal{C}$, then t' is terminal in $\mathcal{C}$.*
- *(Equiv-invariance) Equivalences preserve and reflect terminal objects, i.e., if $F\colon \mathcal{C} \xrightarrow{\simeq} \mathcal{D}$ is an equivalence between strict categories, then an object $t \in \mathcal{C}$ is terminal if and only if $Ft \in \mathcal{D}$ is terminal.*

Proof. We only prove iso-invariance and equiv-invariance. The rest is left as an exercise.

By Corollary 2.6, an isomorphism $t \cong t'$ in $\mathcal{C}$ induces, for every object $x \in \mathcal{C}$, a bijection

$$\mathcal{C}(x,t) \cong \mathcal{C}(x,t').$$

Hence one is a singleton if and only if the other is. This shows that t is terminal if and only if t' is terminal.

Let $F\colon \mathcal{C} \to \mathcal{D}$ be an equivalence and $t \in \mathcal{C}$ be an object. Suppose first that Ft is terminal in $\mathcal{D}$ and let $x \in \mathcal{C}$ be an object. Using that F is fully faithful, we have bijections

$$\mathcal{C}(x,t) \cong \mathcal{D}(Fx,Ft) \cong *.$$

This shows that t is terminal in $\mathcal{C}$.

Now, suppose that t is terminal in $\mathcal{C}$ and let $y \in \mathcal{D}$ be an object. Since F is essentially surjective, there is an object $x \in \mathcal{C}$ together with an isomorphism $y \cong Fx$. Using that F is fully faithful and Corollary 2.6, we obtain bijections

$$* \cong \mathcal{C}(x,t) \cong \mathcal{D}(Fx,Ft) \cong \mathcal{D}(y,Ft).$$

This shows that Ft is terminal in $\mathcal{D}$. □

Initial objects are *dual* to terminal ones in the following sense.

Proposition 4.4. *An object $i \in \mathcal{C}$ is initial in $\mathcal{C}$ if and only if it is terminal in $\mathcal{C}^{\mathrm{op}}$.*

Proof. Exercise. □

By applying Proposition 4.3 to $\mathcal{C}^{\mathrm{op}}$, we obtain the dual results for initial objects.

Corollary 4.5. *Let $\mathcal{C}$ be a strict category.*

- *(Uniqueness) Initial objects are unique up to a unique isomorphism.*
- *(Representability) An object $i \in \mathcal{C}$ is initial if and only if, for every object $x \in \mathcal{C}$, there is a bijection $\mathcal{C}(i,x) \cong *$.*
- *(Iso-invariance) If $i \in \mathcal{C}$ is an initial object and $i \cong i'$ in $\mathcal{C}$, then i' is initial in $\mathcal{C}$.*
- *(Equiv-invariance) Equivalences preserve and reflect initial objects.*

We now prove that the strict category $\mathcal{CAT}_s$ has a terminal and an initial object.

Proposition 4.6. *The strict category $[0]$ is a terminal object in $\mathcal{CAT}_s$ and the empty strict category $\emptyset$ is an initial object in $\mathcal{CAT}_s$.*

Proof. Exercise. □

The notion of terminal object in $\mathcal{CAT}_s$ is equiv-sensitive.

Example 4.7. The functor $0\colon [0] \to \mathcal{I}$ to the walking isomorphism $\mathcal{I} = \{0 \cong 1\}$ is an equivalence of strict categories. However, the strict category $\mathcal{I}$ is not terminal. Indeed, there are two different functors $0, 1\colon [0] \to \mathcal{I}$, selecting the objects 0 and 1, respectively.

The notion of initial object is equiv-invariant, since any strict category that is equivalent to the empty strict category in $\mathcal{CAT}_s$ is itself empty.

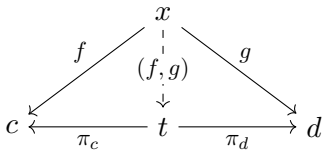
Proposition 4.8. *If $\mathcal{C}$ is equivalent to $\emptyset$, then $\mathcal{C} = \emptyset$.*

Proof. Exercise. □

4.2. **Products and coproducts**

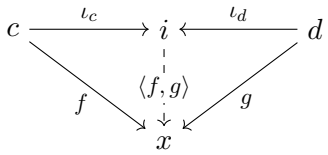
Definition 4.9. Let $\mathcal{C}$ be a strict category and $c, d \in \mathcal{C}$ be objects.

- An object $t \in \mathcal{C}$ is a **product** of c and d if there are morphisms $\pi_c\colon t \to c$ and $\pi_d\colon t \to d$ in $\mathcal{C}$ satisfying the following universal property: for every object $x \in \mathcal{C}$ and all morphisms $f\colon x \to c$ and $g\colon x \to d$ in $\mathcal{C}$, there is a unique morphism $(f,g)\colon x \to t$ such that $\pi_c \circ (f,g) = f$ and $\pi_d \circ (f,g) = g$.



We often write $c \times d$ for a product of c and d. This notation always includes a choice of **canonical projections** $\pi_c \colon c \times d \to c$ and $\pi_d \colon c \times d \to d$.

- An object $i \in \mathcal{C}$ is a **coproduct** of c and d if there are morphisms $\iota_c \colon c \to i$ and $\iota_d \colon d \to i$ in $\mathcal{C}$ satisfying the following universal property: for every object $x \in \mathcal{C}$ and all morphisms $f \colon c \to x$ and $g \colon d \to x$ in $\mathcal{C}$, there is a unique morphism $\langle f, g \rangle \colon i \to x$ such that $\langle f, g \rangle \circ \iota_c = f$ and $\langle f, g \rangle \circ \iota_d = g$.



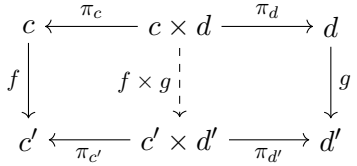
We often write $c \amalg d$ for a coproduct of c and d. This notation always includes a choice of **canonical inclusions** $\iota_c \colon c \to c \amalg d$ and $\iota_d \colon d \to c \amalg d$.

Example 4.10. The strict categories in Example 1.9 have the following (co)products.

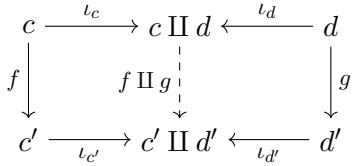
- Given sets S and T, the set $S \times T \coloneqq \{(s, t) \mid s \in S, t \in T\}$ is a product of S and T in $\mathcal{S}et$, and their disjoint union $S \amalg T$ is a coproduct of S and T in $\mathcal{S}et$.
- Given pointed sets (S, s) and (T, t), the pointed set $(S \times T, (s, t))$ is a product of (S, s) and (T, t) in $\mathcal{S}et_\bullet$, and the pointed set $\big((S \amalg T)/(s \sim t), [s]\big)$ is a coproduct of (S, s) and (T, t) in $\mathcal{S}et_\bullet$.
- Given k-vector spaces V and W, their direct sum $V \oplus W$ is both a product and a coproduct of V and W in $\mathcal{V}ect_k$.
- Given topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ in $\mathcal{T}op$, the product $X \times Y$ equipped with the product topology is a product of $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ in $\mathcal{T}op$, and the topological space
$$\big(X \amalg Y, \{U \amalg V \mid U \in \mathcal{T}_X, V \in \mathcal{T}_Y\}\big)$$
is a coproduct of $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ in $\mathcal{T}op$.

Construction 4.11. Let $\mathcal{C}$ be a strict category and $f \colon c \to c'$, $g \colon d \to d'$ be morphisms in $\mathcal{C}$.

- If $c \times d$ and $c' \times d'$ are products of c, d and c', d', respectively, then by the universal property of the product $c' \times d'$, there is a unique morphism $f \times g \colon c \times d \to c' \times d'$ making the following diagram commute.



- If $c \amalg d$ and $c' \amalg d'$ are coproducts of c, d and c', d', respectively, then by the universal property of the coproduct $c \amalg d$, there is a unique morphism $f \amalg g \colon c \amalg d \to c' \amalg d'$ making the following diagram commute.



Proposition 4.12. *Let $\mathcal{C}$ be a strict category and $c, d \in \mathcal{C}$ be objects.*

- *(Uniqueness) Binary products are unique up to a unique isomorphism commuting with the canonical projections, i.e., if $t, t' \in \mathcal{C}$ are products of c and d with canonical projections $\pi_c \colon t \to c$, $\pi_d \colon t \to d$ and $\pi'_c \colon t' \to c$, $\pi'_d \colon t' \to d$, there is a unique isomorphism $\varphi \colon t \cong t'$ such that $\pi'_c \circ \varphi = \pi_c$ and $\pi'_d \circ \varphi = \pi_d$.*
- *(Representability) An object $t \in \mathcal{C}$ is a product of c and d if and only if, for every object $x \in \mathcal{C}$, there is a bijection*
$$\mathcal{C}(x, t) \cong \mathcal{C}(x, c) \times \mathcal{C}(x, d)$$
which is natural in x.
- *(Products with terminal objects) If d is terminal in $\mathcal{C}$, then c is a product of c and d.*
- *(Iso-invariance) If t is a product of c and d in $\mathcal{C}$ and $t \cong t'$ in $\mathcal{C}$, then t' is a product of c and d in $\mathcal{C}$.*
- *(Equiv-invariance) Equivalences preserve and reflect binary products.*

Again coproducts are *dual* to products in the following sense.

Proposition 4.13. *Let $\mathcal{C}$ be a strict category and $c, d \in \mathcal{C}$ be objects. An object $i \in \mathcal{C}$ is a coproduct of c and d in $\mathcal{C}$ if and only if it is a product of c and d in $\mathcal{C}^{\mathrm{op}}$.*

Proof. Exercise. □

By applying Proposition 4.12 to $\mathcal{C}^{\mathrm{op}}$, we obtain the dual results for binary coproducts.

Corollary 4.14. *Let $\mathcal{C}$ be a strict category and $c, d \in \mathcal{C}$ be objects.*

- *(Uniqueness) Binary coproducts are unique up to a unique isomorphism commuting with the canonical inclusions.*
- *(Representability) An object $i \in \mathcal{C}$ is a coproduct of c and d if and only if, for every object $x \in \mathcal{C}$, there is a bijection*
$$\mathcal{C}(i, x) \cong \mathcal{C}(c, x) \times \mathcal{C}(d, x)$$
which is natural in x.
- *(Coproducts with initial objects) If d is initial in $\mathcal{C}$, then c is a coproduct of c and d.*
- *(Iso-invariance) If $i \in \mathcal{C}$ is a coproduct of c and d and $i \cong i'$ in $\mathcal{C}$, then i' is a coproduct of c and d in $\mathcal{C}$.*
- *(Equiv-invariance) Equivalences preserve and reflect binary coproducts.*

We now prove that the strict category $\mathcal{CAT}_s$ has all binary products.

Construction 4.15. Let $\mathcal{C}$ and $\mathcal{D}$ be strict categories. There is a strict category $\mathcal{C} \times \mathcal{D}$ with

- object set $\mathrm{Ob}(\mathcal{C}) \times \mathrm{Ob}(\mathcal{D})$, i.e., objects are pairs (c, d) of objects $c \in \mathcal{C}$ and $d \in \mathcal{D}$;
- for all objects $(c, d), (c', d') \in \mathcal{C} \times \mathcal{D}$, hom set
$$(\mathcal{C} \times \mathcal{D})((c, d), (c', d')) \coloneqq \mathcal{C}(c, c') \times \mathcal{D}(d, d'),$$
i.e., morphisms $(c, d) \to (c', d')$ in $\mathcal{C} \times \mathcal{D}$ are pairs (f, g) of morphisms $f \colon c \to c'$ in $\mathcal{C}$ and $g \colon d \to d'$ in $\mathcal{D}$;
- for each object $(c, d) \in \mathcal{C} \times \mathcal{D}$, the identity morphism $\mathrm{id}_{(c,d)} \coloneqq (\mathrm{id}_c, \mathrm{id}_d) \colon (c, d) \to (c, d)$;
- for all morphisms $(f, g) \colon (c, d) \to (c', d')$ and $(f', g') \colon (c', d') \to (c'', d'')$ in $\mathcal{C} \times \mathcal{D}$, the composite morphism $(f', g') \circ (f, g) \coloneqq (f' \circ f, g' \circ g) \colon (c, d) \to (c'', d'')$.

This is a well-defined strict category. It comes with canonical projections $\pi_{\mathcal{C}} \colon \mathcal{C} \times \mathcal{D} \to \mathcal{C}$ and $\pi_{\mathcal{D}} \colon \mathcal{C} \times \mathcal{D} \to \mathcal{D}$ given by projecting onto the first and second components, respectively.

Proposition 4.16. *Let $\mathcal{C}$ and $\mathcal{D}$ be strict categories. The strict category $\mathcal{C} \times \mathcal{D}$ constructed above is a product of $\mathcal{C}$ and $\mathcal{D}$ in $\mathcal{CAT}_s$.*

Proof. Exercise. □

The construction of products in $\mathcal{CAT}_s$ is equiv-invariant.

Proposition 4.17. *Let $F\colon \mathcal{C} \to \mathcal{C}'$ and $G\colon \mathcal{D} \to \mathcal{D}'$ be equivalences of strict categories. Then the induced functor $F \times G\colon \mathcal{C} \times \mathcal{D} \to \mathcal{C}' \times \mathcal{D}'$ between their products is also an equivalence.*

Proof. Exercise. □

We now prove that the strict category $\mathcal{CAT}_s$ has all binary coproducts.

Construction 4.18. Let $\mathcal{C}$ and $\mathcal{D}$ be strict categories. There is a strict category $\mathcal{C} \amalg \mathcal{D}$ with

- object set $\mathrm{Ob}(\mathcal{C}) \amalg \mathrm{Ob}(\mathcal{D})$;
- for all objects $x, y \in \mathcal{C} \amalg \mathcal{D}$, hom set

$$(\mathcal{C} \amalg \mathcal{D})(x,y) \coloneqq \begin{cases} \mathcal{C}(x,y) & \text{if } x,y \in \mathcal{C} \\ \mathcal{D}(x,y) & \text{if } x,y \in \mathcal{D} \\ \emptyset & \text{else} \end{cases}$$

- identity morphisms those of $\mathcal{C}$ and $\mathcal{D}$;
- composite morphisms those of $\mathcal{C}$ and $\mathcal{D}$.

This is a well-defined strict category. It comes with canonical inclusions $\iota_{\mathcal{C}}\colon \mathcal{C} \to \mathcal{C} \amalg \mathcal{D}$ and $\iota_{\mathcal{D}}\colon \mathcal{D} \to \mathcal{C} \amalg \mathcal{D}$ given by inclusion into the first and second summands, respectively.

Proposition 4.19. *Let $\mathcal{C}$ and $\mathcal{D}$ be strict categories. The strict category $\mathcal{C} \amalg \mathcal{D}$ constructed above is a coproduct of $\mathcal{C}$ and $\mathcal{D}$ in $\mathcal{CAT}_s$.*

Proof. Exercise. □

The construction of coproducts in $\mathcal{CAT}_s$ is equiv-invariant.

Proposition 4.20. *Let $F\colon \mathcal{C} \to \mathcal{C}'$ and $G\colon \mathcal{D} \to \mathcal{D}'$ be equivalences of strict categories. Then the induced functor $F \amalg G\colon \mathcal{C} \amalg \mathcal{D} \to \mathcal{C}' \amalg \mathcal{D}'$ between their coproducts is also an equivalence.*

Proof. Exercise. □

4.3. **Pullback squares**

Definition 4.21. Let $\mathcal{C}$ be a strict category and $f\colon c \to e$ and $g\colon d \to e$ be morphisms in $\mathcal{C}$. An object $t \in \mathcal{C}$ is a **pullback** of f and g if there is a commutative square in $\mathcal{C}$

$$\begin{array}{ccc} t & \xrightarrow{\pi_d} & d \\ {\scriptstyle \pi_c}\downarrow & & \downarrow{\scriptstyle g} \\ c & \xrightarrow[f]{} & e \end{array}$$

satisfying the following universal property: for every object $x \in \mathcal{C}$ and every commutative square in $\mathcal{C}$

$$\begin{array}{ccc} x & \xrightarrow{k} & d \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle g} \\ c & \xrightarrow[f]{} & e \end{array}$$

there is a unique morphism $(h,k)\colon x \to t$ such that $\pi_c \circ (h,k) = h$ and $\pi_d \circ (h,k) = k$.

We often write $c \times_e d$ for a pullback of f and g. This notation always includes a choice of **canonical projections** $\pi_c\colon c \times_e d \to c$ and $\pi_d\colon c \times_e d \to d$.

We often say that a commutative square, as below, is a **pullback square** if it satisfies the universal property of the pullback of f and g, and denote this by decorating the square with the symbol $\lrcorner$ in the top left corner.

$$\begin{array}{ccc} t & \xrightarrow{\pi_d} & d \\ {\scriptstyle \pi_c}\downarrow & \lrcorner & \downarrow{\scriptstyle g} \\ c & \xrightarrow[f]{} & e \end{array}$$

Example 4.22. The strict category $\mathcal{S}et$ has the following pullbacks. Given functions $f\colon S \to U$ and $g\colon T \to U$ of sets, a pullback of f and g in $\mathcal{S}et$ is given by the subset of $S \times T$

$$S \times_U T \coloneqq \{(s,t) \mid f(s) = g(t)\} \subseteq S \times T.$$

Construction 4.23. Let $\mathcal{C}$ be a strict category and consider a commutative diagram in $\mathcal{C}$

$$\begin{array}{ccccc} c & \xrightarrow{f} & e & \xleftarrow{g} & d \\ {\scriptstyle h}\downarrow & & {\scriptstyle l}\downarrow & & \downarrow{\scriptstyle k} \\ c' & \xrightarrow[f']{} & e' & \xleftarrow[g']{} & d' \end{array}$$

If $c \times_e d$ and $c' \times_{e'} d'$ are pullbacks of f, g and f', g', respectively, then by the universal property of the pullback $c' \times_{e'} d'$, there is a unique morphism $h \times_l k\colon c \times_e d \to c' \times_{e'} d'$ making the following diagram commute.

$$\begin{array}{ccccc} c & \xleftarrow{\pi_c} & c \times_e d & \xrightarrow{\pi_d} & d \\ {\scriptstyle h}\downarrow & & {\scriptstyle h \times_l k}\downarrow & & \downarrow{\scriptstyle k} \\ c' & \xleftarrow[\pi_{c'}]{} & c' \times_{e'} d' & \xrightarrow[\pi_{d'}]{} & d' \end{array}$$

Proposition 4.24. *Let $\mathcal{C}$ be a strict category and $f\colon c \to e$ and $g\colon d \to e$ be morphisms in $\mathcal{C}$.*

- *(Uniqueness) Pullbacks are unique up to a unique isomorphism commuting with the canonical projections.*
- *(Representability) An object $t \in \mathcal{C}$ is a pullback of f and g if and only if, for every object $x \in \mathcal{C}$, there is a bijection*

  $$\mathcal{C}(x,t) \cong \mathcal{C}(x,c) \times_{\mathcal{C}(x,e)} \mathcal{C}(x,d)$$

  which is natural in x, where the right-hand side is the pullback of the functions $f_*\colon \mathcal{C}(x,c) \to \mathcal{C}(x,e)$ and $g_*\colon \mathcal{C}(x,d) \to \mathcal{C}(x,e)$.
- *(Products as pullbacks) If e is terminal in $\mathcal{C}$, then an object $t \in \mathcal{C}$ is a pullback of the unique morphisms $f = !\colon c \to e$ and $g = !\colon d \to e$ if and only if it is a product of c and d.*
- *(Iso-invariance) If $t \in \mathcal{C}$ is a pullback of f and g and $t \cong t'$ in $\mathcal{C}$, then t' is a pullback of f and g in $\mathcal{C}$.*
- *(Equiv-invariance) Equivalences preserve and reflect pullbacks.*

We now prove that the strict category $\mathcal{CAT}_s$ has all pullbacks.

Construction 4.25. Let $F\colon \mathcal{C} \to \mathcal{E}$ and $G\colon \mathcal{D} \to \mathcal{E}$ be functors of strict categories. There is a strict category $\mathcal{C} \times_{\mathcal{E}} \mathcal{D}$ with

- object set $\mathrm{Ob}(\mathcal{C}) \times_{\mathrm{Ob}(\mathcal{E})} \mathrm{Ob}(\mathcal{D})$, i.e., objects are pairs (c, d) of objects $c \in \mathcal{C}$ and $d \in \mathcal{D}$ such that $Fc = Gd$ in $\mathcal{E}$;
- for all objects $(c, d), (c', d') \in \mathcal{C} \times_{\mathcal{E}} \mathcal{D}$, hom set
$$(\mathcal{C} \times_{\mathcal{E}} \mathcal{D})((c, d), (c', d')) \coloneqq \mathcal{C}(c, c') \times_{\mathcal{E}(Fc, Fc')} \mathcal{D}(d, d'),$$
i.e., morphisms $(c, d) \to (c', d')$ are pairs (f, g) of morphisms $f\colon c \to c'$ in $\mathcal{C}$ and $g\colon d \to d'$ in $\mathcal{D}$ such that $Ff = Gg$ in $\mathcal{E}$;
- for each object $(c, d) \in \mathcal{C} \times_{\mathcal{E}} \mathcal{D}$, the identity morphism $\mathrm{id}_{(c,d)} \coloneqq (\mathrm{id}_c, \mathrm{id}_d)\colon (c, d) \to (c, d)$;
- for all morphisms $(f, g)\colon (c, d) \to (c', d')$ and $(f', g')\colon (c', d') \to (c'', d'')$ in $\mathcal{C} \times_{\mathcal{E}} \mathcal{D}$, the composite morphism $(f', g') \circ (f, g) \coloneqq (f' \circ f, g' \circ g)\colon (c, d) \to (c'', d'')$.

This is a well-defined strict category. It comes with canonical projections $\pi_{\mathcal{C}}\colon \mathcal{C} \times_{\mathcal{E}} \mathcal{D} \to \mathcal{C}$ and $\pi_{\mathcal{D}}\colon \mathcal{C} \times_{\mathcal{E}} \mathcal{D} \to \mathcal{D}$ given by projecting onto the first and second components, respectively.

Proposition 4.26. *Let $F\colon \mathcal{C} \to \mathcal{E}$ and $G\colon \mathcal{D} \to \mathcal{E}$ be functors of strict categories. The strict category $\mathcal{C} \times_{\mathcal{E}} \mathcal{D}$ constructed above is a pullback of F and G in $\mathcal{CAT}_s$.*

Proof. Exercise. □

Unlike products, the construction of pullbacks in $\mathcal{CAT}_s$ is equiv-sensitive.

Example 4.27. We have the following pullback squares in $\mathcal{CAT}_s$

$$\begin{array}{ccc} [0] & \xrightarrow{\mathrm{id}_{[0]}} & [0] \\ {\scriptstyle \mathrm{id}_{[0]}}\downarrow & \lrcorner & \downarrow{\scriptstyle \mathrm{id}_{[0]}} \\ [0] & \xrightarrow[\mathrm{id}_{[0]}]{} & [0] \end{array} \qquad\qquad \begin{array}{ccc} \emptyset & \xrightarrow{!} & [0] \\ {\scriptstyle !}\downarrow & \lrcorner & \downarrow{\scriptstyle 1} \\ [0] & \xrightarrow[0]{} & \mathcal{I} \end{array}$$

where $\mathcal{I} = \{0 \cong 1\}$ is the walking isomorphism. Moreover, there is a commutative diagram of functors

$$\begin{array}{ccccc} [0] & \xrightarrow{0} & \mathcal{I} & \xleftarrow{1} & [0] \\ {\scriptstyle \mathrm{id}_{[0]}}\downarrow{\scriptstyle \simeq} & & {\scriptstyle !}\downarrow{\scriptstyle \simeq} & & {\scriptstyle \mathrm{id}_{[0]}}\downarrow{\scriptstyle \simeq} \\ [0] & \xrightarrow[\mathrm{id}_{[0]}]{} & [0] & \xleftarrow[\mathrm{id}_{[0]}]{} & [0] \end{array}$$

such that all vertical functors are equivalences, but the induced functor between their pullbacks

$$\emptyset = [0] \times_{\mathcal{I}} [0] \to [0] \times_{[0]} [0] \cong [0]$$

is not an equivalence.

5. Subcategories, localizations, and strict groupoids

We now introduce *strict groupoids*, namely strict categories whose morphisms are all isomorphisms. They generalize sets, regarded as strict categories with only identity morphisms, by allowing objects to be related by isomorphisms and to have nontrivial automorphisms. Thus, a strict groupoid records not only which objects are isomorphic, but also the isomorphisms between them.

In the synthetic setting, the counterparts of the notion of strict groupoids will be basic objects of the theory, playing the role traditionally played by sets. The goal of this section is to construct the strict category of strict groupoids, its canonical inclusion into $\mathcal{CAT}_s$, and both a left and a right adjoint to this inclusion.

5.1. Subcategories

Definition 5.1. A functor $F\colon \mathcal{C} \to \mathcal{D}$ between strict categories is

- **faithful** if, for all objects $x, y \in \mathcal{C}$, the induced function $\mathcal{C}(x,y) \to \mathcal{D}(Fx, Fy)$ is injective;
- **full** if, for all objects $x, y \in \mathcal{C}$, the induced function $\mathcal{C}(x,y) \to \mathcal{D}(Fx, Fy)$ is surjective.

Remark 5.2. Observe that a functor is fully faithful precisely when it is both full and faithful.

Let us fix a strict category $\mathcal{C}$.

Definition 5.3. A functor $\mathcal{D} \to \mathcal{C}$ is a **strict subcategory inclusion**, denoted by $\mathcal{D} \subseteq \mathcal{C}$, if it is injective on objects and faithful. We call $\mathcal{D}$ a **strict subcategory** of $\mathcal{C}$ if there is a strict subcategory inclusion $\mathcal{D} \subseteq \mathcal{C}$.

Remark 5.4. The data of a strict subcategory $\mathcal{D}$ of $\mathcal{C}$ can be given by specifying

- a subset $\mathrm{Ob}(\mathcal{D}) \subseteq \mathrm{Ob}(\mathcal{C})$ of objects;
- for all objects $x, y \in \mathcal{D}$, a subset $\mathcal{D}(x,y) \subseteq \mathcal{C}(x,y)$ of morphisms;

satisfying the conditions:

- identity morphisms are in $\mathcal{D}$: for every object $x \in \mathcal{D}$, the identity id_x is in $\mathcal{D}(x,x)$;
- morphisms in $\mathcal{D}$ are closed under composition: if $f \in \mathcal{D}(x,y)$ and $g \in \mathcal{D}(y,z)$, then their composite $g \circ f$ in $\mathcal{C}$ is in $\mathcal{D}(x,z)$.

The notion of a strict subcategory is equiv-sensitive.

Example 5.5. Recall that the functor $0\colon [0] \to \mathcal{I}$ and the unique functor $\mathcal{I} \to [0]$ are equivalences of strict categories, where $\mathcal{I} = \{0 \cong 1\}$ is the walking isomorphism. The identity functor $\mathrm{id}_{[0]}\colon [0] \to [0]$ is a strict subcategory inclusion, but $\mathcal{I} \xrightarrow{\simeq} [0]$ is not. Moreover, the inclusion $\langle 0, 1\rangle\colon [0] \amalg [0] \to \mathcal{I}$ is a strict subcategory inclusion, but the composite

$$[0] \amalg [0] \to \mathcal{I} \xrightarrow{\simeq} [0]$$

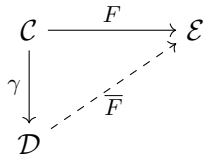
is not.

Definition 5.6. A strict subcategory inclusion $I\colon \mathcal{D} \hookrightarrow \mathcal{C}$ is **full** if the functor I is full. We call $\mathcal{D}$ a **full strict subcategory** of $\mathcal{C}$ if there is a full strict subcategory inclusion $I\colon \mathcal{D} \hookrightarrow \mathcal{C}$.

Remark 5.7. The data of a full strict subcategory $\mathcal{D}$ of $\mathcal{C}$ can be given by specifying a subset $\mathrm{Ob}(\mathcal{D}) \subseteq \mathrm{Ob}(\mathcal{C})$ of objects only. Indeed, we then take the subsets of hom sets consisting of all morphisms and the conditions from Remark 5.4 are then automatically satisfied. We say that $\mathcal{D}$ is the full strict subcategory of $\mathcal{C}$ **spanned by the objects** in $\mathrm{Ob}(\mathcal{D})$.

5.2. Localizations

Definition 5.8. Let $\mathcal{C}$ be a strict category and $W \subseteq \mathcal{CAT}_s([1], \mathcal{C})$ be a subset of morphisms of $\mathcal{C}$. A strict category $\mathcal{D}$ is a **localization** of $\mathcal{C}$ at W if there is a functor $\gamma\colon \mathcal{C} \to \mathcal{D}$ which sends morphisms in W to isomorphisms in $\mathcal{D}$ and satisfies the following universal property: for every strict category $\mathcal{E}$ and every functor $F\colon \mathcal{C} \to \mathcal{E}$ which sends morphisms in W to isomorphisms in $\mathcal{E}$, there is a unique functor $\overline{F}\colon \mathcal{D} \to \mathcal{E}$ such that $\overline{F} \circ \gamma = F$.



We call the functor $\gamma\colon \mathcal{C} \to \mathcal{D}$ the **localization functor**.

We now show that localizations exist by constructing an explicit strict category satisfying the universal property. We may assume that W contains all identity morphisms, since adding them does not change the universal property.

Construction 5.9. Let $\mathcal{C}$ be a strict category and $W \subseteq \mathcal{CAT}_s([1], \mathcal{C})$ be a subset of morphisms of $\mathcal{C}$ containing the identity morphisms and closed under composition, i.e., if $f, g \in W$, then $gf \in W$. There is a strict category $\mathcal{C}[W^{-1}]$ whose

- object set is $\mathrm{Ob}(\mathcal{C})$;
- for all objects $x, y \in \mathcal{C}$, hom set $\mathcal{C}[W^{-1}](x, y)$ is the large set of equivalence classes of finite zigzags
$$x = c_0 \xleftarrow{w_1} c_0' \xrightarrow{f_1} c_1 \xleftarrow{w_2} c_1' \xrightarrow{f_2} \dots \xleftarrow{w_n} c_{n-1}' \xrightarrow{f_n} c_n = y$$
where $n \geq 0$, $w_1, \dots, w_n \in W$ and $f_1, \dots, f_n$ are morphisms in $\mathcal{C}$ modulo the relations:
  - identity morphisms pointing forwards or backwards may be removed or added;
  - adjacent morphisms pointing forwards may be composed in $\mathcal{C}$;
  - adjacent pairs $x \xrightarrow{w} y \xleftarrow{w} x$ or $y \xleftarrow{w} x \xrightarrow{w} y$ with $w \in W$ may be removed;
- identities are given by length 0 zigzags;
- composites are given by concatenation.

This is a well-defined strict category (Exercise).

It comes with a functor $\gamma\colon \mathcal{C} \to \mathcal{C}[W^{-1}]$ which

- acts as the identity on objects;
- sends a morphism $f\colon x \to y$ in $\mathcal{C}$ to the zigzag
$$x \xleftarrow{\mathrm{id}_x} x \xrightarrow{f} y.$$

This is a well-defined functor (Exercise).

Proposition 5.10. *The strict category $\mathcal{C}[W^{-1}]$ together with the functor $\gamma\colon \mathcal{C} \to \mathcal{C}[W^{-1}]$ is a localization of $\mathcal{C}$ at W.*

Proof. Exercise. □

Remark 5.11. Let $\mathcal{C}$ be a strict category and $W \subseteq \mathcal{CAT}_s([1], \mathcal{C})$ be a subset of morphisms of $\mathcal{C}$. Consider the smallest subset $\overline{W} \subseteq \mathcal{CAT}_s([1], \mathcal{C})$ containing W, the identity morphisms of $\mathcal{C}$, and closed under composition. Then the strict category $\mathcal{C}[\overline{W}^{-1}]$ together with the functor $\gamma\colon \mathcal{C} \to \mathcal{C}[\overline{W}^{-1}]$ is a localization of $\mathcal{C}$ at W. This shows that all localizations exist in $\mathcal{CAT}_s$.

Example 5.12. Consider the strict category $\mathcal{C} = \{0 \xrightarrow{s} 1 \xrightarrow{r} 0 \mid rs = \mathrm{id}_0\}$ with two objects 0 and 1, and whose morphisms are id_0, id_1, $s\colon 0 \to 1$, $r\colon 1 \to 0$, and $sr\colon 1 \to 1$, subject to the relation $rs = \mathrm{id}_0$. Then inverting the morphism s in $\mathcal{C}$ yields an isomorphism of strict categories $\mathcal{C}[\{s\}^{-1}] \cong \mathcal{I}$ with the walking isomorphism $\mathcal{I}$. Indeed, in the localization $\mathcal{C}[\{s\}^{-1}]$, we have
$$r = rss^{-1} = s^{-1}$$
since $rs = \mathrm{id}_0$. Thus r and s become mutual inverses.

5.3. **Strict groupoids**

Definition 5.13. A **strict groupoid** is a strict category $\mathcal{G}$ in which every morphism is an isomorphism.

We denote by $\mathcal{G}pd_s$ the full strict subcategory of $\mathcal{C}at_s$ spanned by the small strict groupoids, and by $\mathcal{GPD}_s$ the huge full strict subcategory of $\mathcal{CAT}_s$ spanned by the strict groupoids.

Construction 5.14. The **underlying strict groupoid** of a strict category $\mathcal{C}$ is the strict subcategory of $\mathcal{C}$ given by the strict groupoid $\mathcal{C}^{\simeq}$ whose

- objects are the objects of $\mathcal{C}$;
- morphisms $x \to y$ are the isomorphisms $x \xrightarrow{\cong} y$ in $\mathcal{C}$.

This is a well-defined strict subcategory (Exercise).

Construction 5.15. There is a functor $(-)^{\simeq} \colon \mathcal{C}at_s \to \mathcal{G}pd_s$ sending

- a small strict category $\mathcal{C}$ to its underlying strict groupoid $\mathcal{C}^{\simeq}$;
- a functor $F \colon \mathcal{C} \to \mathcal{D}$ in $\mathcal{C}at_s$ to the induced functor $F^{\simeq} \colon \mathcal{C}^{\simeq} \to \mathcal{D}^{\simeq}$ obtained by restricting F to underlying strict groupoids. Since F preserves isomorphisms, this is well-defined.

This is a well-defined functor (Exercise).

Proposition 5.16. *The inclusion $\iota \colon \mathcal{G}pd_s \subseteq \mathcal{C}at_s$ and the underlying strict groupoid functor form an adjunction*

$$\iota \colon \mathcal{G}pd_s \rightleftarrows \mathcal{C}at_s \colon (-)^{\simeq}.$$

Proof. Exercise. □

Proposition 5.17. *Let $\mathcal{C}$ be a strict category and $W \coloneqq \mathcal{CAT}_s([1], \mathcal{C})$ be the set of all morphisms in $\mathcal{C}$. The localization $\Pi\mathcal{C} \coloneqq \mathcal{C}[W^{-1}]$ of $\mathcal{C}$ at W is a strict groupoid, called the **groupoidification** of $\mathcal{C}$. We denote its localization functor by $\gamma_{\mathcal{C}} \colon \mathcal{C} \to \Pi\mathcal{C}$.*

Proof. Considering the construction $\mathcal{C}[W^{-1}]$ from before, it suffices to prove that each zigzag is an isomorphism. However, this follows from taking the reversed zigzag with morphisms pointing in the opposite direction. □

Construction 5.18. There is a functor $\Pi \colon \mathcal{C}at_s \to \mathcal{G}pd_s$ sending

- a small strict category $\mathcal{C}$ to its groupoidification $\Pi\mathcal{C}$;
- a functor $F \colon \mathcal{C} \to \mathcal{D}$ in $\mathcal{C}at_s$ to the unique functor $\Pi F \coloneqq \overline{\gamma_{\mathcal{D}} \circ F} \colon \Pi\mathcal{C} \to \Pi\mathcal{D}$

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{F} & \mathcal{D} \\ {\scriptstyle \gamma_{\mathcal{C}}}\downarrow & & \downarrow{\scriptstyle \gamma_{\mathcal{D}}} \\ \Pi\mathcal{C} & \xrightarrow[\overline{\gamma_{\mathcal{D}} \circ F}]{} & \Pi\mathcal{D} \end{array}$$

  given by the universal property of the localization $\gamma_{\mathcal{C}} \colon \mathcal{C} \to \Pi\mathcal{C}$, using that the composite $\gamma_{\mathcal{D}} \circ F \colon \mathcal{C} \to \Pi\mathcal{D}$ sends all morphisms in $\mathcal{C}$ to isomorphisms in $\Pi\mathcal{D}$.

This is a well-defined functor (Exercise).

Proposition 5.19. *The groupoidification functor and the inclusion $\iota \colon \mathcal{G}pd_s \subseteq \mathcal{C}at_s$ form an adjunction*

$$\Pi \colon \mathcal{C}at_s \rightleftarrows \mathcal{G}pd_s \colon \iota.$$

Proof. Exercise. Prove this using the universal property of the groupoidification, as a localization at all morphisms. □

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